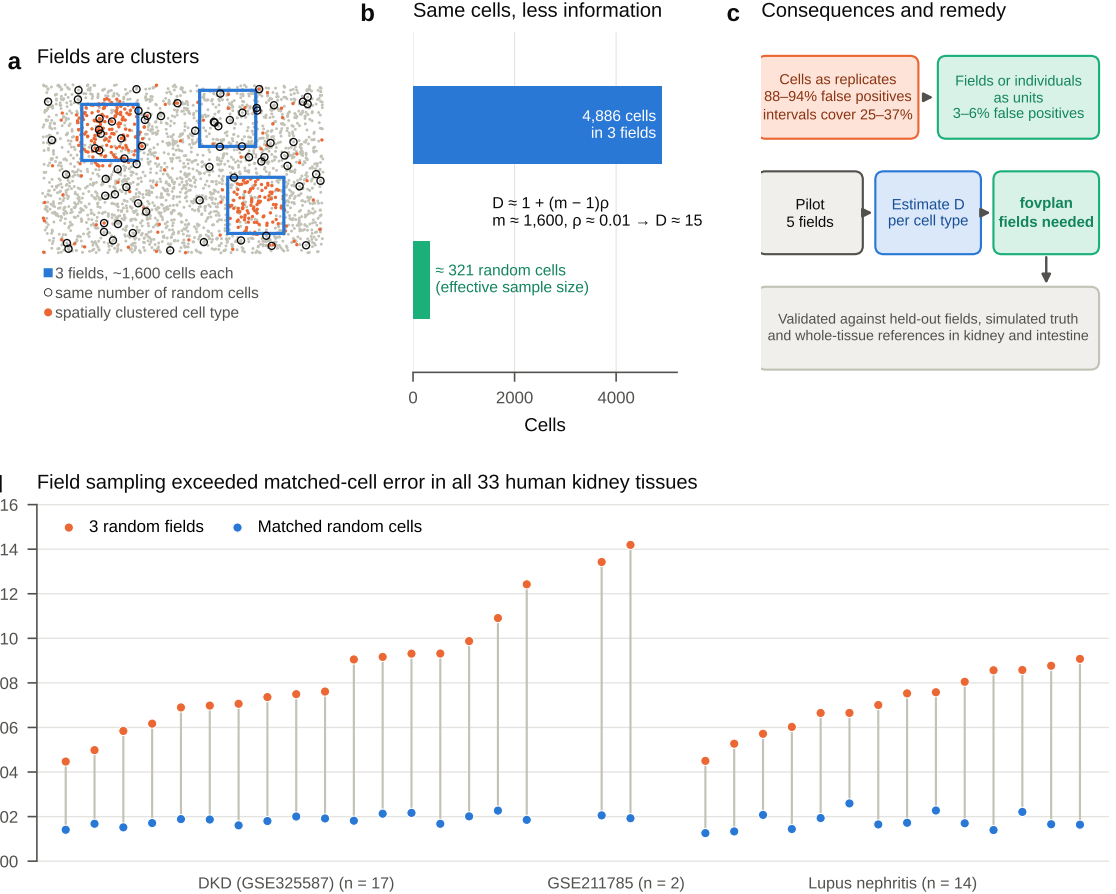

a
Fields are clusters
3 fields, ~1,600 cells each
same number of random cells
spatially clustered cell type
b
Same cells, less information
4,886 cells
in 3 fields
D ≈ 1 + (m − 1)ρ
m ≈ 1,600, ρ ≈ 0.01 → D ≈ 15
≈ 321 random cells
(effective sample size)
0
2000
4000
Cells
c
Consequences and remedy
Cells as replicates
88–94% false positives
intervals cover 25–37%
Fields or individuals
as units
3–6% false positives
Pilot
5 fields
Estimate D
per cell type
fovplan
fields needed
Validated against held-out fields, simulated truth
and whole-tissue references in kidney and intestine
d
Field sampling exceeded matched-cell error in all 33 human kidney tissues
3 random fields
Matched random cells
Median composition
error (TV)
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
DKD (GSE325587) (n = 17)
GSE211785 (n = 2)
Lupus nephritis (n = 14)